



Full System Test Document

UI components · runtime systems · platform validation

Document ID	MV-FST-1.0
Product	MooreVIEW PC / Cloud Studio
Platforms	MVP Suite (Windows appliance) · IoT-Link (Linux) · Cloud SaaS
Generated	2026-07-23

How to use this document

This checklist validates **every major UI surface** and **backend system** before release, customer handoff, or regression after a large merge.

Mark each line: ☐ PASS · ☒ FAIL · ☐ N/A (feature disabled or not licensed)

Tip: Press **F1** in-app for contextual Help. Press **F2** for Training. Use **Project** → **System setup** for feature toggles and project load.

Recommended order

- 1. Run automated baseline (Section 1).
- 2. Complete **Platform A** (MVP Suite) Sections 2–12.
- 3. Repeat Sections 2–12 on **Platform B** (Cloud SaaS) where applicable.
- 4. Complete sign-off (Section 13).

Default test credentials

Platform	URL	Login
MVP Suite appliance	http://127.0.0.1:3090	admin@local / ChangeMeAdmin! or operator@local
Cloud SaaS (after seed)	http://127.0.0.1:3100	Org demo, operator@demo.local / demo
Cloud admin	same	admin@demo.local / ChangeMeAdmin!

Load a known project (e.g. duplex lift station) before HMI/runtime tests.

1. Automated baseline (run first)

```
cd est-pc
npm test
npm run test:baseline
curl -s http://127.0.0.1:3090/health
curl -s -o /dev/null -w "%{http_code}" http://127.0.0.1:3090/api/sys-log
```

Check	Expected
npm test	All tests pass
npm run test:baseline	ST logic + tag/HMI checks pass
/health	HTTP 200, JSON ok
/api/sys-log (authenticated)	HTTP 200, not 403 cellular gate

2. Authentication & shell

2.1 Login / session

Login

- └─ [] Appliance login page loads (/login)
- └─ [] Valid credentials → redirect to dashboard (/)
- └─ [] Invalid credentials → error message, no session
- └─ [] Logout clears session; protected routes redirect to login
- └─ [] Session persists across page refresh

Cloud SaaS only

- └─ [] Organization ID field required
- └─ [] Tenant slug resolves; wrong org rejected
- └─ [] Cloud Studio top nav (Sites, Fleet, People) visible when entitled

2.2 Dashboard shell

Main layout

- └─ [] Top bar renders (logo, menus, status strip)
- └─ [] HMI canvas loads selected screen
- └─ [] Tag values update (live or simulated)
- └─ [] Alarm banner / horn state reflects active alarms
- └─ [] Footer / status text shows project name and mode
- └─ [] No console errors on initial load (DevTools)

3. Top bar menus

3.1 Project ▾

Project

- └─ [] Open project... (file picker / hub)
- └─ [] Save project
- └─ [] Save project as...
- └─ [] Deploy project... (cloud catalog)
- └─ [] Share project... (cloud catalog)
- └─ [] Recent projects list opens known project
- └─ [] Export / print project configuration
- └─ [] System setup... (opens setup popup)
- └─ [] Project hub... (local / cloud / file tabs)
- └─ [] Delete project...
- └─ [] Reload runtime
- └─ [] Exit / close behavior (platform appropriate)

3.2 Historian ▾

Historian

- └─ [] Historian... (main historian popup)
- └─ [] Historian logger...
- └─ [] Shortcuts match Help inventory

3.3 Reporting ▾

Reporting

- └─ [] Reports... (report popup, all sections scroll)
- └─ [] No duplicate CMMS menu item (CMMS is top-bar only)
- └─ [] Mongo logs / hardware history load without 403

3.4 Tools ▾

Tools

- └─ [] Program editor...
- └─ [] Tag editor...
- └─ [] Driver setup...
- └─ [] Camera admin...
- └─ [] MV Draw (opens /mv-draw or embedded)
- └─ [] IO map (when project provides map)
- └─ [] Connectivity
 - └─ [] Cellular SIMs (/cellular/sims) – if entitled
 - └─ [] Cloud Sims (/cloud/sims) – if entitled
- └─ [] Training... (F2)

3.5 Help ▾

Help

- └─ [] Help... (F1) – topics load, search works
- └─ [] About / version info
- └─ [] External doc links open (if configured)

3.6 Top bar direct actions

Direct buttons

- └─ [] Alarms... (alarm popup)
- └─ [] CMMS (→ /cmms when feature enabled)
- └─ [] Camera quick access (if enabled)
- └─ [] User / account menu (cloud or appliance auth UI)

4. Popups — Program & tags

4.1 Program editor popup

Program

- └─ [] Popup opens from Tools → Program editor
- └─ [] Program list loads; select program shows ST source
- └─ [] Deploy / reload actions succeed (or show clear error)
- └─ [] Version / CRC display matches device when Parc connected
- └─ [] Close popup returns to dashboard without layout break

4.2 Tag editor popup

Tags

- └─ [] Tag tree or list loads
- └─ [] Filter / search finds known tag
- └─ [] Edit tag metadata (where allowed) saves
- └─ [] Force write / simulate value (if exposed) updates HMI
- └─ [] Refresh reloads from server

5. Popups — Alarms & drivers

5.1 Alarms popup

Alarms

- └─ [] Active alarm list matches runtime state
- └─ [] Acknowledge single alarm
- └─ [] Acknowledge all (if available)
- └─ [] Alarm history / shelve (if enabled)
- └─ [] Horn silence / reset behaves per project logic

5.2 Driver setup popup

Drivers – tabs

- └─ [] Driver list – list loads, enable/disable driver
- └─ [] Modbus RTU ↔ TCP – mapping UI loads
- └─ [] NextCentury API – config panel loads
- └─ [] Save persists after popup close + reopen
- └─ [] MQTT Parc / Opta driver status visible when configured

6. Popups — Cameras

6.1 Camera admin (all tabs)

Cameras

- └─ [] Overview – summary counts, health
- └─ [] Inventory – add/edit/remove camera entries
- └─ [] Detail – stream preview or placeholder
- └─ [] Archive – recordings list (if go2rtc/archive enabled)
- └─ [] Events – motion/AI events when configured
- └─ [] AI history – AI detection log
- └─ [] Discover – ONVIF/network scan
- └─ [] System – go2rtc / bridge status
- └─ [] Settings – credentials, retention, messages

7. Popups — Historian & logging

7.1 Historian popup

Historian

- └─ [] Tag picker / trend selection
- └─ [] Chart renders with time range
- └─ [] Pan/zoom or range presets work
- └─ [] Export CSV or image (if available)
- └─ [] Maintenance... panel opens (retention, DB tools)

7.2 Historian logger popup

Historian logger

- └─ ☐ Logger tag list loads
- └─ ☐ Enable/disable logging per tag or group
- └─ ☐ Sample interval / deadband fields save
- └─ ☐ Data appears in historian after logging interval

7.3 Historian maintenance panel

Maintenance

- └─ ☐ Mongo / store connection status shown
- └─ ☐ Purge or compact actions (if exposed) require confirm
- └─ ☐ No permission errors for admin role

8. Popups — Reports

8.1 Reports popup sections

Reports

- └─ ☐ System log (Mongo) – entries load, filter works
 - └─ ☐ Hardware history – entries load
 - └─ ☐ ROI panel – edit assumptions, Save ROI
 - └─ ☐ Open CMMS link → navigates to /cmms
 - └─ ☐ Print / export sections (where implemented)
 - └─ ☐ Refresh buttons update without stale-cache errors
-

9. Popups — System setup

9.1 Setup tabs

```
System setup
├─ General
│   ├── [ ] Site name, timezone, display options save
│   ├── [ ] Load on boot / startup project modes
│   ├── [ ] MQTT Parc hub + cloud remote uplink sections
│   └── [ ] Alarm notification users under Alarms popup (not General tab)
├─ Features
│   ├── [ ] User access matrix (cmms, cameras, cellular, cloud sims...)
│   ├── [ ] Toggle off CMMS hides top-bar CMMS link after reload
│   └── [ ] Operator role respects gates (403 or hidden UI)
├─ Area configuration (hardware)
│   ├── [ ] ROI / areas list loads
│   └── [ ] Save updates HMI bindings
└─ Projects
    ├── [ ] Active project path displayed
    ├── [ ] Print project configuration...
    └── [ ] Switch bundled / local project loads HMI

Alarms popup (related)
├─ CMMS / MQTT integration
│   ├── [ ] External CMMS MQTT publish settings (separate from /cmms UI)
│   └── [ ] Save persists; distinct from integrated work orders
```

9.2 Project hub popup

```
Project hub
├─ [ ] Local repository – list, open, clone
├─ [ ] MV Cloud – sync / pull / push (when cloud hub configured)
└─ [ ] File – upload .est / archive, extract, open
```

10. Popups — Training & Help

Training (F2)

- └─ [] Section tabs populate from training manifest (M0-M15, CBM-1-13)
- └─ [] M9 lab covers integrated /cmms (not MQTT-only)
- └─ [] M15 lab references F1 → Cameras & video
- └─ [] Quizzes load and score
- └─ [] Close returns to dashboard

Help (F1)

- └─ [] Topic tree includes Integrated CMMS, Cameras & video, Cloud Sims
- └─ [] Screen layout lists top-bar CMMS (not Reporting → CMMS)
- └─ [] System setup documents Features access matrix
- └─ [] In-app search finds "CMMS", "Cameras", "Historian", "MQTT"
- └─ [] Keyboard F1 opens from dashboard focus

11. Standalone pages

Standalone routes

- └─ /cmms
 - └─ [] Page loads (not 404)
 - └─ [] Overview – KPI cards, open WOs
 - └─ [] Work orders – create, assign, complete
 - └─ [] Preventive maintenance – schedules list
 - └─ [] Alarm-created WO appears after runtime alarm (integrated mode)
- └─ /mv-draw
 - └─ [] Canvas editor loads
 - └─ [] Save / open drawing
 - └─ [] Help panel in MV Draw
- └─ /io-map (project-specific)
 - └─ [] PDF or SVG map displays
 - └─ [] Point labels match tag names
- └─ /cellular/sims
 - └─ [] Gated off when feature disabled (403 or redirect)
 - └─ [] SIM inventory CRUD when enabled
- └─ /cloud/sims
 - └─ [] Virtual device lab loads
 - └─ [] Sim lifecycle actions work
- └─ Cloud-only: /sites, /sites/devices, /fleet, /people
 - └─ [] Each page loads for entitled tenant
 - └─ [] Pairing / fleet map / user admin per CLOUD_USER_GUIDE.md

12. HMI & runtime systems

12.1 HMI canvas

HMI

- └─ [] Screen navigation (buttons, layers, composites)
- └─ [] Animations / indicators follow tag values
- └─ [] Faceplate popups (pumps, valves, lifts) open and command
- └─ [] Write confirmation or interlock messages appear
- └─ [] Duplex lift indicators cycle correctly (if project loaded)
- └─ [] Assisted living / demo screens (if bundled) bind correctly

12.2 Runtime & MQTT Parc

Runtime

- └─ [] Scan engine running (tags updating)
- └─ [] ST program executes (outputs follow logic)
- └─ [] MQTT Parc hub connects when broker + device online
- └─ [] Program deploy match/version shown in UI
- └─ [] Float mirror / duplex mirror tags consistent
- └─ [] Reload runtime recovers after project change

12.3 Persistence & project

Project system

- └─ [] Save writes .est / workspace files
- └─ [] Open project restores tags, HMI, drivers
- └─ [] Archive pack/unpack (workspace.est.zip) if used
- └─ [] Startup loader picks configured default project

12.4 CMMS integration (system)

CMMS backend

- └─ [] GET /api/cmms/dashboard returns counts
- └─ [] GET /api/cmms/assignees lists tenant/appliance users
- └─ [] POST work order creates record
- └─ [] Alarm bridge creates WO on new alarm (when enabled)
- └─ [] External MQTT CMMS publish (setup) independent of /cmms UI

12.5 Auth & feature gates

Feature gates

[]

requireFeature middleware returns 403 for disabled API

[]

UI elements hidden when feature false in appliance_auth / tenant

[]

CMMS, cameras, cellular, cloud sims independently gated

[]

Operator vs admin capability difference verified

13. Platform matrix

Area	MVP Suite :3090	IoT-Link Linux	Cloud SaaS :3100
Appliance login	Yes	Yes	Tenant login
Full dashboard	Yes	Yes	Yes
CMMS /cmms	If entitled	If entitled	If entitled
Cellular sims	Optional	Optional	Usually N/A
Cloud sims	Optional	Optional	Often enabled
Project hub cloud tab	If hub URL set	If hub URL set	Native
Sites / Fleet / People	N/A	N/A	Cloud nav
Portable / SaaS bundle scripts	Windows scripts	deploy/iot-link	deploy/cloud

14. Regression hotspots

After changes in these areas, re-run the marked sections:

Change location	Re-test sections
public/js/help.js, public/js/training.js	10
public/js/app.js, API fetch	2, 8, 12.5
cellularSims.js, cloudSims.js routes	1, 3.4, 8.1
CMMS (cmmsApp.js, alarm bridge)	3.6, 8.1, 11, 12.4
HMI / hmi.js, SVG composites	12.1
MQTT Parc / Opta driver	5.2, 12.2
Feature catalog / auth	9.1, 12.5
Cloud Studio UI	11 (cloud pages), 13

15. Sign-off

Field	MVP Suite	Cloud SaaS	Tester initials
Test date			
Build / git SHA			
Project under test			
Sections 2–12 complete	☐	☐	
Automated baseline (§1) pass	☐	☐	
Blocking defects			
Approved for release	☐	☐	

Approver signature: _____ **Date:** _____

MooreVIEW Full System Test · MV-FST-1.0 · Help/training aligned 2026-07 · See also
`docs/testing/CMMS_USER_TESTING.md` and `docs/BASELINE_TEST.md` .